Educated to Be Mothers: Indoctrination and Demographic Backlash

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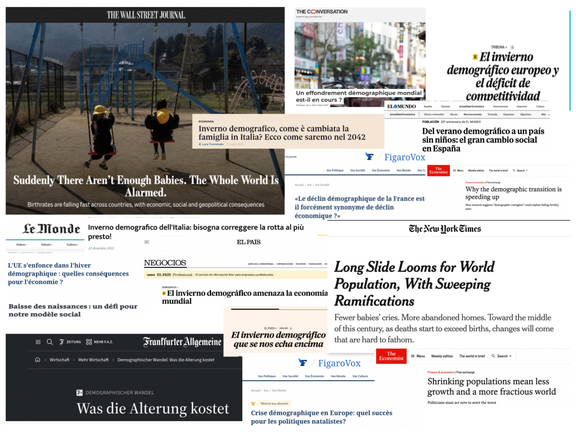
Motivation



Demographic Winter (i.e., *declining birth rates coupled with aging population*) has become a salient issue among national governments and public opinions due to its potential negative effect on:

- Innovation, productivity and economic growth slowdown
- Welfare system sustainability
- Cultural and societal shifts (backward?)

Western democracies have started to address some of its economic aspects:

- The cost of childbearing: provision of financial and/or fiscal incentives depending on the number of children (e.g., Family Act 2020, Social Security Contribution Exemption with 2+ children in Budget Law 2024)
- Work-life balance: maternity/paternity leaves, provision of affordable kindergartens (e.g., Kindergarten Bonus in Budget Law 2024)

Part of the electorate view demographic winter as primarily due to the change in gender roles within households, advocating for a return to a "traditional family" via:

- Indirect intervention by rising the economic costs of alternative models (e.g., lack of universal childcare system in the US)
- Direct intervention on beliefs (e.g., propaganda, indoctrination)

"Respect for stay-at-home mothers has been poisoned by a toxic combination of the village elders' war on the traditional family and radical feminism's misogynistic crusade to make working outside the home the only marker of social value and self-respect"
US Senate's third-ranking Republican senator **Rick Santorum** (*It Takes a Family*, 2005)

What we do

We study how direct intervention on beliefs about gender roles affects childbearing decisions
↓
RQ: What is the impact of ***indoctrination*** *on* ***fertility***

- We leverage a policy reform in the Primary Education System imposed by the Francoist regime in Spain
  - Gender-specific classes
  - Explicit aim to educate girls *"to be mothers (and wives)"*
- We exploit two sources of variation in exposure to the reform deriving from:
  1. Relative size of birth-cohorts
  2. Proportion of teachers cleansed by the Francoist regime
- Novel historical dataset

What we find

In principle, the impact of indoctrination on fertility is not obvious *a priori*:

- Indoctrination works → fertility ↑
- Indoctrination does not work → fertility =
- Indoctrination creates backlash → fertility ↓

- Fertility dropped significantly in places with a bigger share of population potentially exposed to the reform and where the implementation of the reform was more intense.
- The drop in fertility for treated cohorts is associated with a shift in beliefs on gender roles and motherhood, back-lashing the indoctrination perpetrated by the regime.

Literature Review

- Cultural beliefs, the marriage market and fertility: Chong and La Ferrara (2009), La Ferrara et al. (2012)
- Indoctrination
  - Autocratic regimes: Waldinger (2010), Adena et al. (2015), Voigtländer and Voth (2015), Cantoni et al. (2017), Homola et al. (2020), Meriläinen and Mitrunen (WP2024)
    - **Spain**: Balcells and Villamil (2020), Martinez-Marquina (2024)
    - **Gender roles and Female employment**: Gonzales et al. (WP 2024)
  - Democracies: Masella and Clots-Figueras (2008, 2013), Fuchs-Schündeln and Masella (2016), Fouka (2020), Arold (2024)
- Family policies on fertility: Baudin and Stelter (WP 2023)

Historical Background

Civil War (1936-1939)
Primary Education Law under Franco rule *(BOE 1945)*

- Religion became compulsory at all educational levels
- Strict control of all educational materials and contents (Canales, 2021)
- Prohibition of regional languages (e.g., Basque, Catalan, Galician) and mandatory use of the Spanish language
- Gender-specific education
  - Separate classes for boys and girls
  - Different education for girls to prepare for domestic life and motherhood
- Teachers were subordinated to the national interest

Cleansing of Teachers

- The purge of teachers stood among the cleansing of other professionals: teachers were perceived as a threat to the traditional order.
- Three phases:
  1. Killings during the first months of the war
  2. Disorganized purges
  3. Creation of provincial commissions in charge of the examination of teachers
     1. Formed by a high school principal, primary school inspector, president of the association of parents and 2 people of max influence and moral and technical worthiness
     2. They requested at least four different reports from the mayor, the military police, the parish priest, and a person of absolute ideological reliability
- Historians estimate 60,000 teachers were depurated. Approx. 25% of teachers were sanctioned

Data

Panel data for 56 years (1930-1985). 100 units of observation: capital of province and rest of province

- Population censuses from 1930 to 1970: number of inhabitants by age and gender
- Fertility and weddings: Natural movement of the population - National Institute of Statistics *(INE)*
- Teachers (44k files): General Archive of the Administration
- Channels: Center for Sociological Research *(CIS)* individual 1979-92 surveys
- Additional variables: pre-dictatorial elections (1931), unofficial political census from the Francoist regime (1947), distance to the Civil War front

Measuring Fertility:

- *Total Fertility Rates* are the most suited to make comparisons within and between units. Unfortunately, we do not have the number of childbirths by women's age groups.
- Solution: *General Fertility Rates*.
  - Number of children in each capital/rest of province divided by number of women in fertile age, i.e., aged 15-44, in 1940
  - Legitimate measure as long as the size (and composition) of fertile age groups remains constant in time and space
  - This number does not significantly vary between treated and control units
  - Therefore, this allows us to rule out the possible effect of the reform on the population structure

Isolating the effect of the reform:

- Lack of data on fertility rates by age group does not allow us to directly estimate the effect of the reform on fertility.
  - We cannot directly compare, within and between units, the fertility rate of age groups exposed or not to the reform over time
  - We can compare the general fertility rate within and between units where the share of population exposed to the reform is higher or lower
- Solution: Exploiting the cleansing of teachers as a further source of variation in the intensity of the reform within the same group of cohorts and between units

Empirical strategy: DiD

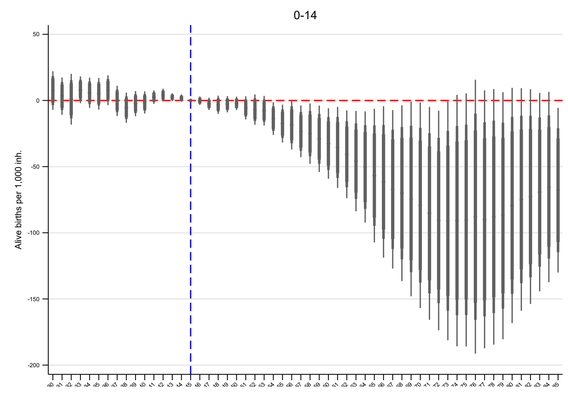
$$y_{it} = \beta_0 + \beta_1(Post_t \times T1_i \times T2_i) + \mathbf{X}_{it} + \Gamma_i + \tau_t + \epsilon_{it} \quad (1)$$

	(1) Coefficient	(2) SD
Religious communities (1930)	0.010	0.052
Professed (pc, 1930)	0.486	0.920
Novices (pc,1930)	0.133	0.405
Lay (pc, 1930)	-0.075	0.202
Republicans (%, 1931)	-0.316	3.172
Socialists (%, 1931)	0.409	2.075
Monarchic (%, 1931)	2.913	3.396
Comunsits (%, 1931)	0.023	0.101
Addicted to regime (%, 1947)	-1.080	3.330
National schools (1000s,1942)	0.559***	0.203
Primary teachers (1000s,1942)	0.281	0.209
Enrolled students (1000s,1942)	-0.698	8.209
Area in the war front (%)	0.115	5.166
Women aged 15-24 (%, 1940)	0.006	0.011
Women aged 25-34 (%, 1940)	-0.001	0.007
Women aged 35-44 (%, 1940)	-0.005	0.006

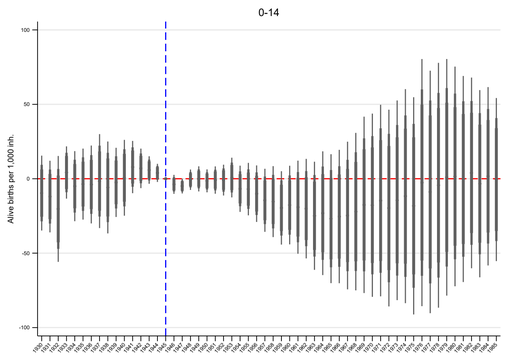
- i denotes an administrative unit and t indicates year
- y_{it} = General Fertility Rate = $\frac{\#births}{\text{Women 15-44 in 1940}}$
- $Post = 1\{> 1945\}$
- $T1_i$ is the female population share with 0-14 years old in 1940
- $T2_i$ is the number of teachers cleansed per 10,000 inhabitants
- $\mathbf{X}_{it} = Post_t \times T1_i, Post_t \times T2_i, Post_t \times MalePop^{0-14}_{i,1940}, Post_t \times MalePop^{0-14}_{i,1940} \times T2_i$

Main results

	(1) Births	(2) Births
Treat (W) X Post 1945	-20.922***	-43.511***
	(5.924)	(14.836)
Treat (M) X Post 1945		23.023*
		(11.859)
Observations	4,591	4,591
R-squared	0.566	0.581
Mun. FE	YES	YES
Year FE	YES	YES
Cohorts	0-14	0-14
Mean of depvar	116.9	116.9

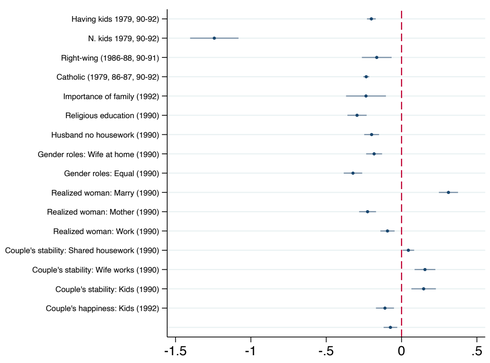


Robustness checks



- Trends
- Discrete treatment
- Changing denominator of GFR
- Placebo: Population 15-44 in 1940
- **Placebo: Population 0-14 in 1930**
- Weddings
- Excluding demographic transition

Mechanisms



We estimate this equation using the sample of female respondents:

$$y_i = \beta_0 + \beta_1 Cohorts\ 1931\text{-}1940_i + \beta_2 Cohorts\ 1941\text{-}1970_i + \Gamma_i + \epsilon_i$$

- y_i are respondent i's demographics and beliefs about religion, family and gender roles
- *Cohorts 1931-1940*$_i$ is a dummy equal to 1 for cohorts only partially exposed to the reform
- *Cohorts 1941-1970*$_i$ is a dummy equal to 1 for cohorts fully exposed to the reform
- Γ_i are fixed effects at the municipality level
- ϵ_i are heteroskedastic-robust standard errors clustered at the locality level
- Some outcome variables (i.e., kids, ideology and religion) are observed in multiple waves, allowing us to control for year fixed effects

10. Discussion and Conclusions

- This study contributes to the literature on the demographic effects of political indoctrination and education policies
- The Francoist regime's primary education reform led to significant demographic changes
- The policy, which included the cleansing of teachers and indoctrination through education, generated unintended effects on the fertility and marriage rates in Spain
- Analysis of beliefs shows lasting hostility in the following decades by the cohorts exposed to Franco's education reforms towards the values of religion, family and gender roles promoted by the regime
- This hostility was likely a key driver of the fertility backlash documented here